

Study

Forest Plot: Comparative Effectiveness of Dynamic/Complex Models vs. Baselines
Effect Size = Relative Improvement in Prediction Performance (%)

Effect [95% CI]

KEY FINDINGS:

- 5/6 studies (83%) favor intervention
- Effect range: -2.3% to +28.9%
- Largest effects: Falls (+18.8%), Progression (+28.9%)
- Only 2/6 test dynamic vs. static hypothesis
- *95% CI estimated (not reported in original)

Comparison Type

- Dynamic vs Static (n=2)
- Static vs Static (n=4)

HETEROGENEITY BARRIERS:
X Different metrics (IAUC, AUC, sMAPE, F-measure, Accuracy)
X Missing variance estimates (0/6 reported intervention CI)
X Different prediction goals (Progression, Falls, Cognitive)
X Different horizons (6-36 months)
△ Meta-analysis NOT POSSIBLE

Ren et al.

Chaithanya et al. 2025

van Wegen et al. 2016

Gao et al. 2018

Tan et al. 2022

Pishva et al. 2022†

-40 -30 -20 -10 0 10 20 30 40 50

Relative Improvement in Performance (%)

Favors Comparator
(Simpler Model Better)

Favors Intervention
(Complex Model Better)

† Did not meet strict inclusion criteria

* 95% confidence intervals estimated (not reported in original studies)

Effect size = Relative improvement of intervention vs. comparator

Positive values favor intervention (more complex model); negative values favor comparator

+9.3% [+5.7, +12.9]*

+28.9% [+17.6, +40.2]*

+18.8% [+0.1, +0.2]

-2.3% [-3.2, -1.4]*

+4.3% [+2.6, +6.0]*

+2.7, +6.1]*